

Fall 2025.

Physics 509. Quantum Theory I.
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Homework # 5.
Due to December 1st, 2025.

You can use any source to solve the problems. However, your solutions should be presented in a very detailed and logical way. You need to understand every step of your solution and be able to explain it.

1. Determine energy levels of a particle in the following potentials using the WKB approximation:

a)
$$U(x) = \begin{cases} \infty, & x < 0 \\ 0, & 0 < x < l \\ \infty, & x > l \end{cases};$$

b)
$$U(x) = U_0 |x/a|;$$

c)
$$U(x) = -U_0 \cosh^{-2} x/a.$$

Hint: use the Bohr-Sommerfeld quantization rule.

2. Use the WKB approximation to calculate transmission coefficients for the following potential barriers:

a)
$$U(x) = \begin{cases} 0, & x < 0 \\ U_0, & 0 < x < l \\ 0, & x > l \end{cases}$$

b)
$$U(x) = U_0 e^{-|x/a|};$$

c)
$$U(x) = \begin{cases} 0, & x < 0 \\ U_0 x/l, & 0 < x < l. \\ 0, & x > l \end{cases}$$

Comment: for the case c) we will be able to calculate the answer up to some preexponential factor. The latter will appear because of the violation of the quasiclassical approximation for the vertical wall of the potential barrier.